

# Homework 02

## Question 1: Training and test error under a fixed design

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Consider a linear regression problem with  $n = 100$  observations and  $p = 20$  available covariates. Generate one matrix  $\mathbf{X}_{\text{all}} \in \mathbb{R}^{n \times p}$  whose entries are independent  $\mathcal{N}(0, 1)$  random variables, and then keep this realized matrix fixed. Let

$$\beta_j = 0.4^{\sqrt{j}}, \quad j = 1, \dots, p,$$

and define

$$\boldsymbol{\mu} = \mathbf{X}_{\text{all}} \boldsymbol{\beta}.$$

Generate one training response and one independent test response from

$$\begin{aligned} \mathbf{y} &= \boldsymbol{\mu} + \boldsymbol{\epsilon}, \\ \mathbf{y}^* &= \boldsymbol{\mu} + \boldsymbol{\epsilon}^*, \end{aligned}$$

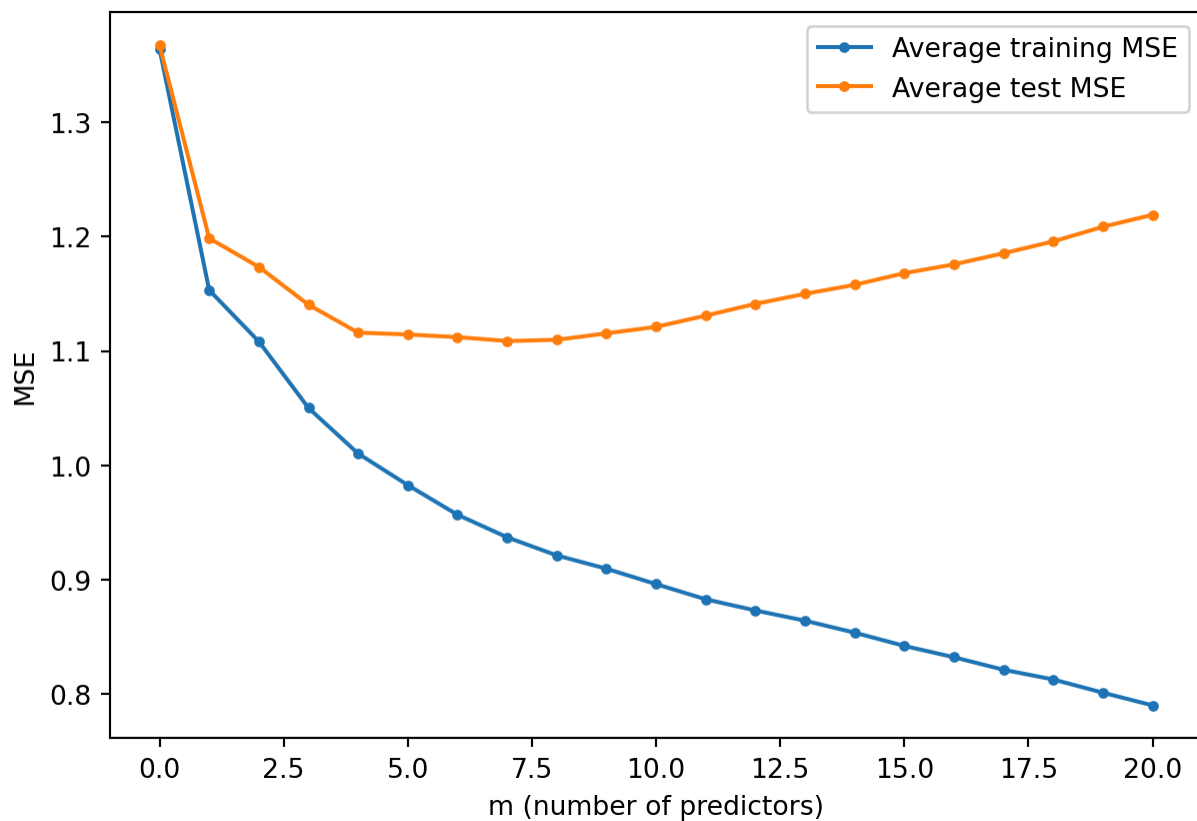
where  $\boldsymbol{\epsilon}, \boldsymbol{\epsilon}^* \stackrel{\text{ind}}{\sim} \mathcal{N}_n(\mathbf{0}, \mathbf{I}_n)$ . Repeat this process independently 200 times, keeping  $\mathbf{X}_{\text{all}}$  fixed.

- For each  $m$ , calculate the average training and test MSE over the 200 simulation runs. Plot both curves against  $m$ . Verify numerically that training MSE does not increase as predictors are added in each simulation run.

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Max increase in training MSE across all runs/steps: -2.3765502898953628e-09

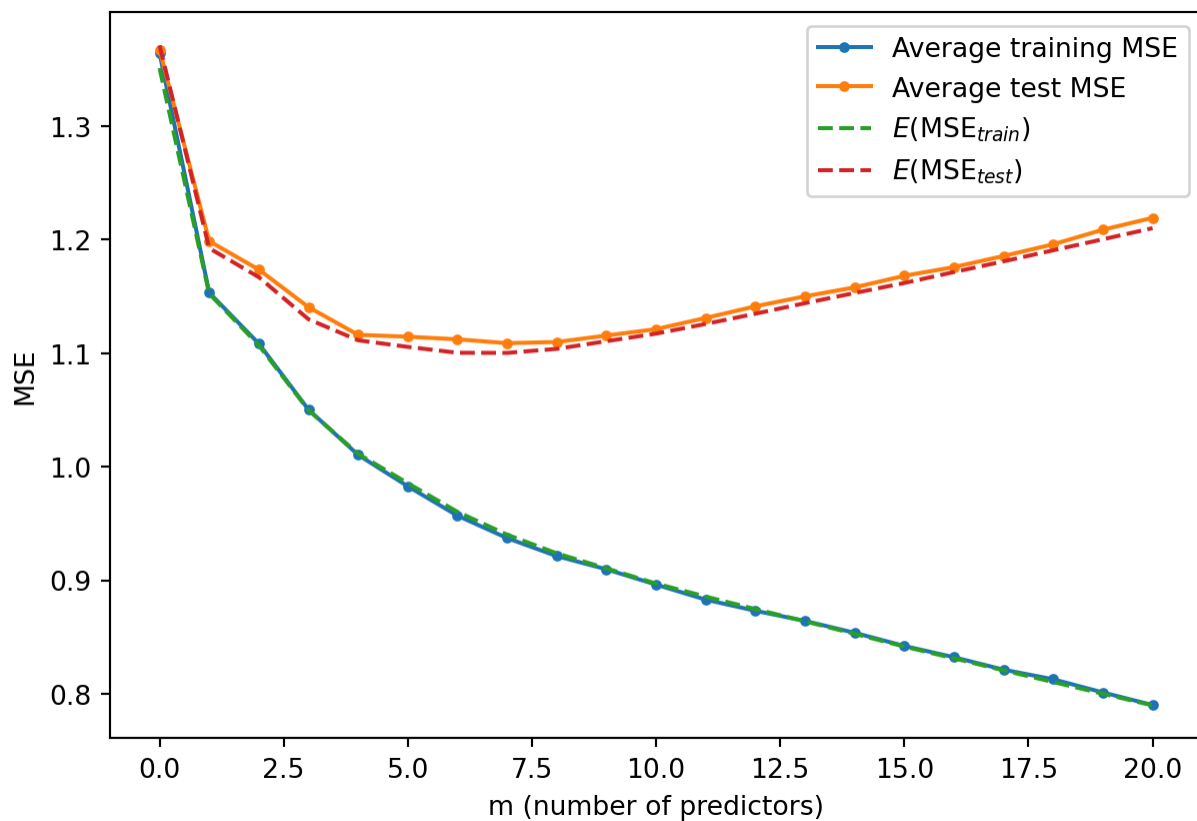
Training MSE non-increasing in every run: True



Training MSE never increases (max change  $\leq 0$  up to rounding): enlarging  $\mathbf{X}_m$  enlarges its column space, so the least-squares RSS for the same  $\mathbf{y}$  cannot increase.

- b. For the model containing the first  $m$  predictors, write  $\mathbf{X}_m = [\mathbf{1}, \mathbf{x}_1, \dots, \mathbf{x}_m]$  and  $\mathbf{H}_m = \mathbf{X}_m(\mathbf{X}_m^\top \mathbf{X}_m)^{-1} \mathbf{X}_m^\top$ . Define the total squared approximation bias  $B_m^2 = \|(\mathbf{I}_n - \mathbf{H}_m)\boldsymbol{\mu}\|_2^2$ . Calculate the two theoretical expectations and add them to your plot.

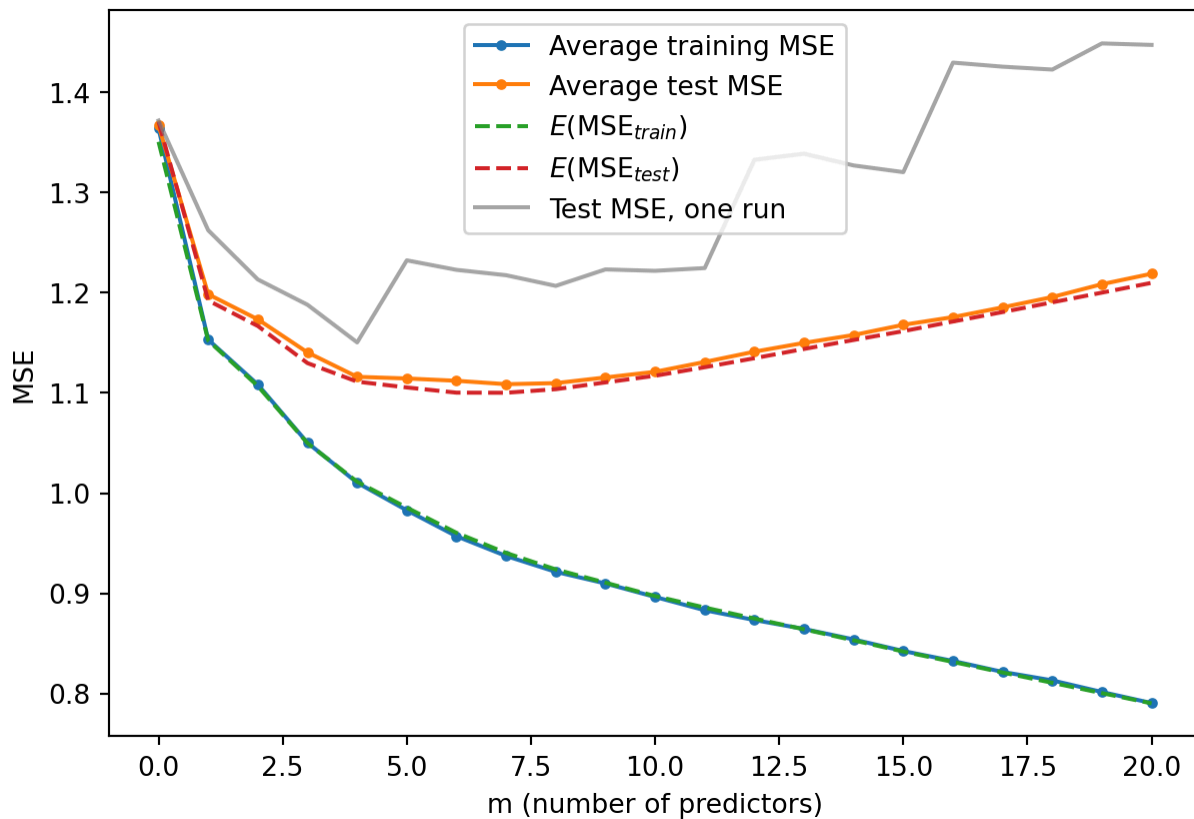
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The theoretical curves track the simulation averages closely at every  $m$ , as expected: averaging 200 independent draws of  $\epsilon$ ,  $\epsilon^*$  converges to the conditional expectations by the law of large numbers.

- c. Add the test MSE curve from one simulation run to the plot. Explain why this single curve is less smooth than the average test MSE, and why training MSE alone cannot be used to select the predictor count.

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The single-run curve is noisy because it reflects only one draw of  $\epsilon, \epsilon^*$ ; averaging over 200 runs cancels that noise. Training MSE always decreases in  $m$  (the  $-(m+1)/n$  term keeps falling even after bias stops shrinking), so it rewards fitting noise rather than a better fit to  $\mu$ . Test MSE instead adds  $+(m+1)/n$ , trading bias against variance, so it is minimized at an interior  $m$ .

## Question 2: Prediction error at one target point

Let  $n = 100$ ,  $p = 6$ , and  $\sigma^2 = 1$ . Construct the fixed covariate matrix  $\mathbf{X}_{\text{all}} \in \mathbb{R}^{n \times p}$  with entries

$$(\mathbf{X}_{\text{all}})_{ij} = \sqrt{2} \cos \left\{ \frac{\pi j(i - \frac{1}{2})}{n} \right\}.$$

Keep  $\mathbf{X}_{\text{all}}$  fixed and generate one training response from  $\mathbf{y} = \mathbf{X}_{\text{all}}\beta + \epsilon$ ,  $\beta = (0.5, \dots, 0.5)^\top$ . Consider prediction at  $\mathbf{x}_0 = (0, 1, 0, 0, 1, 0)^\top$ , with  $\mu_0 = \mathbf{x}_0^\top \beta$ .

- Calculate  $\mu_0$ , the theoretical squared bias, and the expected squared error for  $m = 0, \dots, p$ . Identify where the expected squared error changes and where it remains unchanged.

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```
mu0 = 1.0
m=0: bias^2=1.0000, variance term=0.0100, expected sq. error=1.0100
m=1: bias^2=1.0000, variance term=0.0100, expected sq. error=1.0100
m=2: bias^2=0.2500, variance term=0.0200, expected sq. error=0.2700
m=3: bias^2=0.2500, variance term=0.0200, expected sq. error=0.2700
```

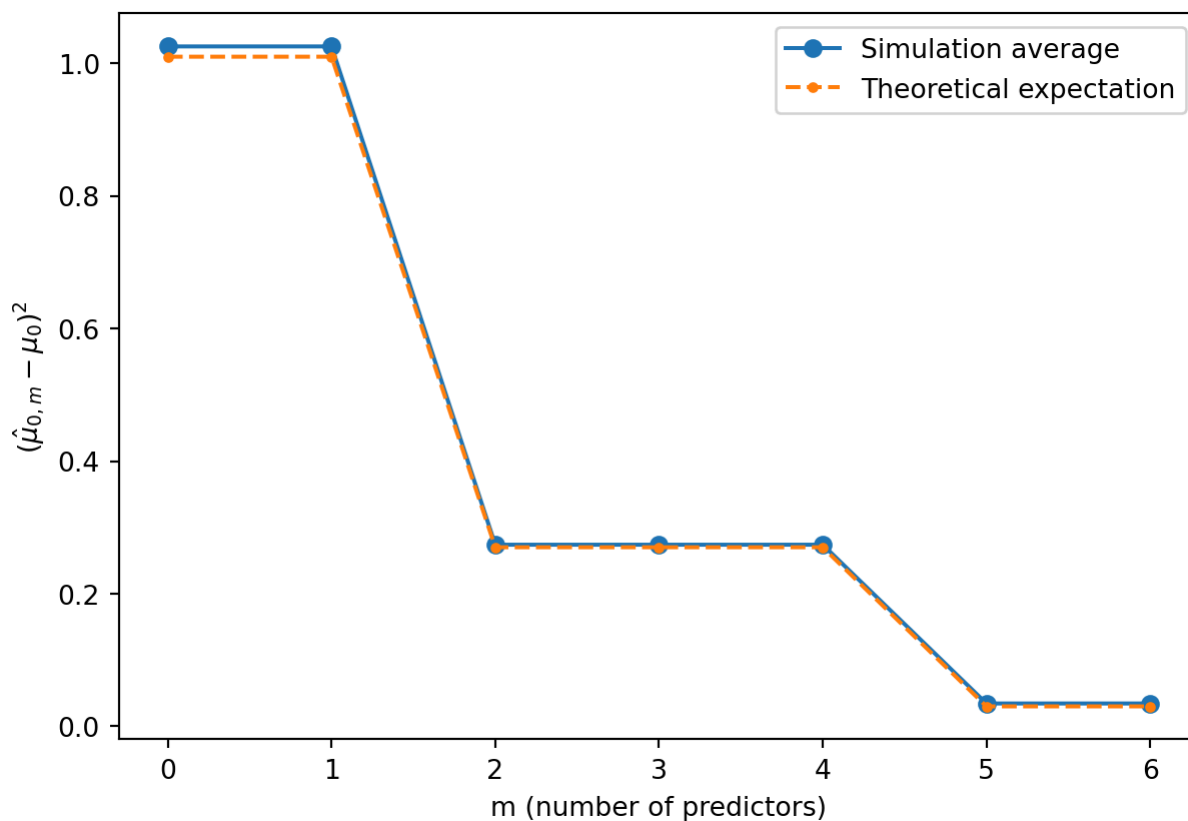
m=4: bias<sup>2</sup>=0.2500, variance term=0.0200, expected sq. error=0.2700  
 m=5: bias<sup>2</sup>=0.0000, variance term=0.0300, expected sq. error=0.0300  
 m=6: bias<sup>2</sup>=0.0000, variance term=0.0300, expected sq. error=0.0300  
 Changes at m → m+1: [ 0.   -0.74  0.    0.   -0.24  0. ]

Only  $x_{0,2} = 1$  and  $x_{0,5} = 1$  are nonzero, so both the bias and variance terms change only when  $m$  crosses 2 or 5 — dropping/rising at  $m = 1 \rightarrow 2$  and  $m = 4 \rightarrow 5$ , flat elsewhere. The expected error is a squared bias from omitted mean-response components plus a variance from estimating each included coefficient.

- b. Use seed 43203. Generate the response, fit the models, and calculate  $(\hat{\mu}_{0,m} - \mu_0)^2$ , repeated 200 times. Plot the simulation averages and theoretical expectations together. Report the smallest  $m$  that minimizes the theoretical error.

► Show the starter code

Simulation-average squared error: [1.0255 1.0255 0.2739 0.2739 0.2739 0.0344 0.0344]  
 Theoretical expected squared error: [1.01 1.01 0.27 0.27 0.27 0.03 0.03]  
 Smallest m minimizing theoretical error: 5



The simulation averages track the theoretical curve closely. Both are minimized at  $m = 5$ : by then both nonzero-loading directions are captured, and adding predictor 6 only adds variance with no bias reduction, so  $m = 6$  just ties  $m = 5$ .

- c. Explain why only predictors 2 and 5 contribute directly to the mean response at  $\mathbf{x}_0$ . Contrast this target-specific error with the test MSE averaged over all rows of  $\mathbf{X}_{\text{all}}$  in Question 1.

$\mu_0 = \sum_j x_{0j}\beta_j$  weights each coefficient by  $x_{0j}$ , and  $x_{0j} = 0$  for  $j \in \{1, 3, 4, 6\}$ , so those coefficients don't enter  $\mu_0$  even though they're nonzero. Question 1's test MSE instead averages error over all rows of  $\mathbf{X}_{\text{all}}$ , so it reflects every predictor's contribution — a predictor irrelevant at one target point can still lower overall test MSE by helping elsewhere.

### Question 3: The optimism correction

Choose one candidate model before observing the response, with design  $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$ , hat matrix  $\mathbf{H}$ ,  $\mathbf{y} = \boldsymbol{\mu} + \boldsymbol{\epsilon}$ ,  $\mathbf{y}^* = \boldsymbol{\mu} + \boldsymbol{\epsilon}^*$  independent given  $\mathbf{X}$ , each with mean  $\mathbf{0}$  and covariance  $\sigma^2 \mathbf{I}_n$ .

a. Derive  $E(\text{MSE}_{\text{train}} \mid \mathbf{X})$  and  $E(\text{MSE}_{\text{test}} \mid \mathbf{X})$ .

**Training MSE.** Since  $\mathbf{H}\mathbf{y} = \mathbf{H}\boldsymbol{\mu} + \mathbf{H}\boldsymbol{\epsilon}$ ,

$$\mathbf{y} - \mathbf{H}\mathbf{y} = (\mathbf{I} - \mathbf{H})\boldsymbol{\mu} + (\mathbf{I} - \mathbf{H})\boldsymbol{\epsilon}.$$

Using  $(\mathbf{I} - \mathbf{H})$  symmetric and idempotent,

$$\|\mathbf{y} - \mathbf{H}\mathbf{y}\|^2 = \|(\mathbf{I} - \mathbf{H})\boldsymbol{\mu}\|^2 + 2\boldsymbol{\mu}^\top (\mathbf{I} - \mathbf{H})\boldsymbol{\epsilon} + \boldsymbol{\epsilon}^\top (\mathbf{I} - \mathbf{H})\boldsymbol{\epsilon}.$$

The cross term vanishes since  $E(\boldsymbol{\epsilon} \mid \mathbf{X}) = \mathbf{0}$ , and

$E[\boldsymbol{\epsilon}^\top (\mathbf{I} - \mathbf{H})\boldsymbol{\epsilon} \mid \mathbf{X}] = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{H}) = \sigma^2(n - (p + 1))$ . Dividing by  $n$ ,

$$E(\text{MSE}_{\text{train}} \mid \mathbf{X}) = \frac{B^2}{n} + \sigma^2 \left(1 - \frac{p + 1}{n}\right).$$

**Test MSE.** Since  $\mathbf{y}^* - \mathbf{H}\mathbf{y} = (\mathbf{I} - \mathbf{H})\boldsymbol{\mu} + \boldsymbol{\epsilon}^* - \mathbf{H}\boldsymbol{\epsilon}$ , with  $\boldsymbol{\epsilon}^*$  independent of  $\boldsymbol{\epsilon}$  and both mean zero, every cross term vanishes in expectation, leaving

$$E(\text{MSE}_{\text{test}} \mid \mathbf{X}) = \frac{1}{n} \left[ B^2 + E(\boldsymbol{\epsilon}^{*\top} \boldsymbol{\epsilon}^* \mid \mathbf{X}) + E(\boldsymbol{\epsilon}^\top \mathbf{H}^\top \mathbf{H} \boldsymbol{\epsilon} \mid \mathbf{X}) \right].$$

Here  $E(\boldsymbol{\epsilon}^{*\top} \boldsymbol{\epsilon}^* \mid \mathbf{X}) = \sigma^2 n$ , and since  $\mathbf{H}^\top \mathbf{H} = \mathbf{H}$ ,  $E(\boldsymbol{\epsilon}^\top \mathbf{H} \boldsymbol{\epsilon} \mid \mathbf{X}) = \sigma^2 \text{tr}(\mathbf{H}) = \sigma^2(p + 1)$ . Hence

$$E(\text{MSE}_{\text{test}} \mid \mathbf{X}) = \frac{B^2}{n} + \sigma^2 \left(1 + \frac{p + 1}{n}\right).$$

b. Deduce the expected optimism and evaluate it for  $n = 120$ ,  $p = 4$ ,  $B^2/n = 0.04$ ,  $\sigma^2 = 1$ .

The expected optimism is

$$E(\text{MSE}_{\text{test}} - \text{MSE}_{\text{train}} \mid \mathbf{X}) = \frac{2\sigma^2(p + 1)}{n}.$$

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```
E(MSE_train | X) = 0.9983333333333334
E(MSE_test | X) = 1.0816666666666668
```

```
Expected optimism = 0.08333333333333337
2*sigma^2*(p+1)/n = 0.08333333333333333
```

This ordering holds only in expectation — for a single realized pair, sampling variability can still make  $\text{MSE}_{\text{test}} < \text{MSE}_{\text{train}}$ .

c. Calculate  $\widehat{\text{MSE}}_{\text{test}}$  and  $C_p$  for  $\text{RSS} = 116.4$ ,  $\hat{\sigma}^2 = 0.96$ ,  $n = 120$ ,  $p = 4$ , and relate them.

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```
MSE_test_hat = 1.05
C_p = 11.2500000000000014
sigma2_hat * (1 + C_p / n) = 1.0500000000000003
```

Solving  $C_p = \text{RSS}/\hat{\sigma}^2 - n + 2(p + 1)$  for RSS and substituting gives

$$\widehat{\text{MSE}}_{\text{test}} = \hat{\sigma}^2 \left( 1 + \frac{C_p}{n} \right),$$

confirmed numerically above. This is a strictly increasing function of  $C_p$  when  $n, \hat{\sigma}^2$  are shared across candidates, so minimizing  $C_p$  always minimizes  $\widehat{\text{MSE}}_{\text{test}}$ .

## Question 4: Comparing Mallows' $C_p$ , AIC, and BIC

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```
Full model: p = 10  RSS = 1089451.721  df = 359  sigma2_hat = 3034.68446
Model A:      p = 7  RSS = 1098908.566
Model B:      p = 8  RSS = 1089717.057
```

a.  $p = 10$  for the full reference model, with residual  $\text{RSS} \approx 1089451.72$ , residual degrees of freedom  $n - (p + 1) = 370 - 11 = 359$ , and  $\hat{\sigma}^2 = \text{RSS}_{\text{full}}/359 \approx 3034.6845$ . Model A has  $p = 7$  with  $\text{RSS}_A \approx 1098908.57$ ; Model B has  $p = 8$  with  $\text{RSS}_B \approx 1089717.06$ .

b.

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```
Model A: Cp=8.116, AIC_red=2974.640, BIC_red=3005.948
Model B: Cp=7.087, AIC_red=2973.532, BIC_red=3008.754
Cp selects: B
AIC selects: B
BIC selects: A
```

$C_p$  and reduced AIC select Model B; reduced BIC selects Model A.

c.

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Reduction in RSS from adding  $s_6$ : 9191.5089  
 Cp penalty per parameter: 2  $\Rightarrow$  2  
 AIC penalty per parameter: 2  $\Rightarrow$  2  
 BIC penalty per parameter:  $\log(n) = 5.9135$

Adding  $s_6$  reduces RSS only slightly.  $C_p$  and AIC charge a flat 2-unit penalty per parameter, small enough that the improvement clears it — both favor Model B. BIC charges  $\log(n) \approx 5.91$  per parameter, too large for the same improvement to clear, so it prefers Model A. Selecting a model this way only shows the best fit-complexity trade-off among the candidates considered, not that it's the true data-generating model.

## Question 5: Validation and final test data

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```
Test MSE by m: [5822.89  5527.791 5527.21  3719.235 3246.823 3234.919 3207.615 2636.502
 2624.924 2369.344 2484.776]
Selected m: 9   reported 'final' MSE: 2369.344
```

- a. The misuse begins at step 2: computing test MSE for all eleven models already uses the test data to compare candidates. Step 3 compounds this by picking the model that scored lowest by chance. Taking the minimum of eleven noisy estimates is systematically optimistic — the test data both chooses the model and grades it, so the reported error understates the true future error.
- b.

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```
5-fold CV MSE by m: [5986.262 5812.74  5875.806 3913.162 3698.52  3751.63  3751.973
 3259.964
 3270.345 3147.616 3121.674]
CV-selected m: 10
Final test MSE at CV-selected m: 2484.776
```

Split the training data (rows 1-370) into five folds; for each  $m$ , fit on four folds and validate on the fifth, cycling through all five, and average to get a CV MSE for that  $m$ . Select  $m$  minimizing CV MSE — no test data involved. Refit that model on all training data, then evaluate once, at the end, on the final test data (rows 371-442).

- c. The eleven values can still be reported as an exploratory summary of how test error varied with  $m$ , but not as an unbiased estimate of future prediction error, since the selection step in part (a) already biases it downward. A valid final estimate needs a fresh test set that played no role in choosing  $m$ .

## Reference

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James, Witten, Hastie, Tibshirani, and Taylor, [\*An Introduction to Statistical Learning\*](#), Chapters 3, 5, and 6.